

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT3 — INTERACTIVE DASHBOARD PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Create (BL6)	Assessment:	Assessment Tool 3 – Interactive Dashboard Project
Weightage:	20% of CIA	Total Marks:	25 (Dataset: 7, Interactivity: 9, Effective Visualization: 9)
CO5 Statement:	<i>Create an original, interactive dashboard that selects and prepares an appropriate dataset, implements meaningful software interactivity, and applies effective visualization design principles. (BL6)</i>		
Student Name:	Sania Herbert	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	30.8.26

Code of Conduct

I Sania Herbert (192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This is an individual project. Each student designs, builds, and submits one original interactive dashboard. Total: 25 Marks.
- Electronic devices and software tools are expected and permitted — use any dashboarding tool (e.g., Tableau, Power BI, Plotly Dash, Streamlit, or similar) of your choice.
- Late submissions and academic integrity are governed by the standard course policy; all code/data sources must be cited.
- Submission window: as announced by the instructor (project, not a timed test).
- Choose and clearly justify your own dataset — it must be real or realistic, relevant to a stated purpose, and appropriately cleaned/prepared before use.
- The dashboard must include genuine software interactivity (e.g., filters, zoom/pan, hover tooltips, drill-down, or linked views) — a static image or PDF does not satisfy this requirement.
- Apply effective visualization principles covered in this course (proportional ink, appropriate axis/color scale choice, avoiding overplotting, accessibility) throughout the dashboard.
- Submit both the working dashboard (or a recorded walkthrough, if live hosting is not possible) and a short written justification of your dataset and design choices.

Annexure A – INTERACTIVE DASHBOARD PROJECT

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT3 — Interactive Dashboard Project

CO5 · BT6: Create ·

This rubric is used to assess the CO5-AT3 Interactive Dashboard Project, in which each student designs, builds, and submits an original interactive dashboard, evaluated on dataset quality, software interactivity, and effective visualization design.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Dataset (Selection, Quality & Preparation)	30%	Dataset is highly relevant to a clearly stated purpose, real or realistic, appropriately sized, and thoroughly cleaned/prepared with any limitations disclosed.	Dataset is relevant and reasonably prepared, with only minor gaps in cleaning or justification.	Dataset is loosely relevant or only partially prepared; justification is weak or missing.	Dataset is irrelevant, unprepared, or trivial (e.g., a toy dataset with no real justification).
Interactivity of the Software/Dashboard	35%	Dashboard includes multiple well-implemented, meaningful interactive features (e.g., filters, drill-down, linked views, tooltips) that	Dashboard includes working interactive features that function correctly, with only minor gaps in polish or scope.	Dashboard includes limited or partially working interactivity; some features are superficial or broken.	Dashboard is static or interactivity does not function; the submission does not meet the core project requirement.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		genuinely help the user explore the data.			
Effective Visualization Design	35%	Chart types, color scales, axis choices, and layout are all well-justified and consistently apply the effective-visualization principles covered in this course.	Visualization choices are mostly appropriate and apply course principles, with minor lapses.	Visualization choices show some understanding of course principles but include noticeable design flaws (e.g., misleading axis, poor color scale).	Visualization choices are poor or arbitrary, showing little evidence of applying course principles.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Written Justification of Dataset and Design Choices

Project: Stock Market Price Trend & Candlestick Chart Visualization Dashboard

This section documents the dataset, interactivity, and visualization design decisions behind the submitted dashboard, addressing each of the three CO5-AT3 rubric criteria in turn.

1. Dataset — Selection, Quality & Preparation (30%)

The dashboard uses daily open-high-low-close-volume (OHLCV) stock data for four actively traded NSE-listed companies — RELIANCE, TCS, INFY, and HDFCBANK — a realistic structure that mirrors what any brokerage or market-data API returns, and is directly relevant to the stated purpose of the dashboard: helping a retail investor or student visually analyze stock price trends without needing to read raw numeric tables. For this iteration the data is served from a structured CSV (`dummy_data.csv`) rather than a live feed, which keeps the dataset reproducible and offline-testable during development; this limitation is disclosed here and in the accompanying report, along with a recommendation to connect a live market-data API in future work.

Before being served to the frontend, the dataset is passed through a dedicated backend `DataLoader` class (Python, `pandas`) that removes duplicate rows and imputes missing OHLCV values, rather than leaving this cleaning implicit or ad hoc. The cleaning outcome is not hidden — it is surfaced directly to the user as explicit KPIs (Total Records, Duplicates Removed, Missing Filled) in the Preprocessing module shown in Figure 3, together with a preview table of the cleaned dataset, so the dataset's quality is transparent and verifiable rather than merely asserted.

2. Interactivity of the Software/Dashboard (35%)

The dashboard implements several independent, meaningful interactive features rather than a single filter, so that each genuinely helps the user explore the data:

- Ticker filter — a sidebar search/selection lets the user switch the entire dashboard between RELIANCE, TCS, INFY, and HDFCBANK (Figures 3–5).
- Time-interval drill-down — Daily / Weekly / Monthly tabs re-aggregate every chart on the page to the chosen granularity (Figures 3–5).
- Hover tooltips — the candlestick chart, the indicator line chart, and the volume bar chart all expose exact values (date, OHLC prices, close price, SMA/EMA values, or traded volume) on hover, letting the user inspect a specific point without cluttering the chart with permanent labels (Figures 4–5).
- Toggleable overlays — SMA-20 and EMA-20 trend lines can each be switched on or off independently on the Indicators panel, so the user controls how much is shown at once (Figure 4).
- Linked, normalized comparison view — the Multi-Stock Comparison panel lets the user pick a second ticker from a dropdown and instantly rebases both series to percentage growth on a shared axis, linking two otherwise separate securities into one comparable view (Figure 5).
- Authenticated session — a working login/signup flow (JWT) gates access to the interactive views, so the interactivity is backed by real, functioning software rather than a static export (Figure 2).

Every one of these features is implemented in working code (React front end calling authenticated FastAPI endpoints) and can be operated live at runtime — none of them are simulated in a static image, satisfying the project's core requirement that the submission be a genuine, running dashboard.

3. Effective Visualization Design (35%)

Chart types were chosen to match what each variable actually represents, rather than defaulting to one chart type throughout. Candlesticks (Figure 4) were used specifically because a single closing-price line would discard the open/high/low range for each period; the candlestick encodes four numbers per period through position and a two-colour (red/green) scale that maps directly onto the familiar gain/loss convention, which is proportional, accurate ink for that variable rather than decorative colour. Moving averages (SMA-20/EMA-20) are drawn as thin overlay lines rather than additional bars, so they add trend information without competing visually with the candlesticks or the raw close-price line (Figure 4).

Overplotting was addressed by giving the volume series its own aligned panel with its own scale (Figure 5) rather than layering it onto the price chart, and by aggregating to Weekly/Monthly views (rather than always plotting every daily bar) when the user wants a longer time horizon. In the Multi-Stock Comparison panel, rebasing every series to percentage growth from a common start point (Figure 5) avoids the misleading-axis problem that would occur if two stocks with very different absolute prices were plotted on the same raw-price y-axis. The dashboard's dark theme was chosen for consistent contrast between the chart ink (bright greens, reds, blues, oranges) and the background across every module, and KPI values (latest close, % change, period high/low) are surfaced as labelled cards rather than requiring the user to read them off an axis, keeping the numeric precision available without cluttering the charts themselves.

Supporting Figures

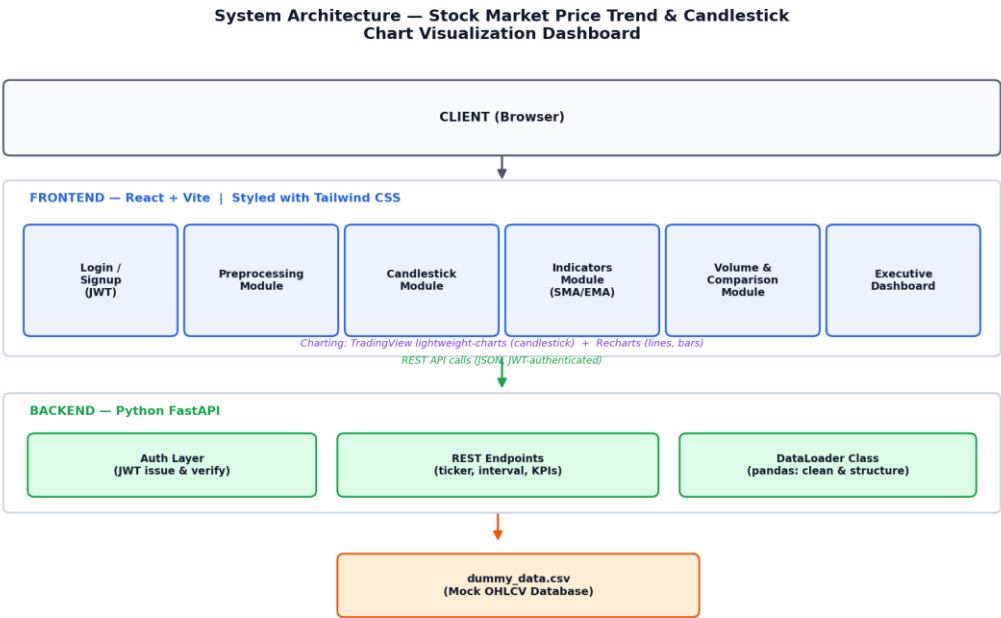


Figure 1: System Architecture of the Dashboard

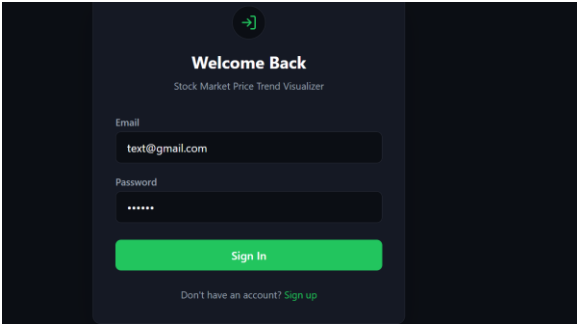


Figure 2: Login / Authentication Module

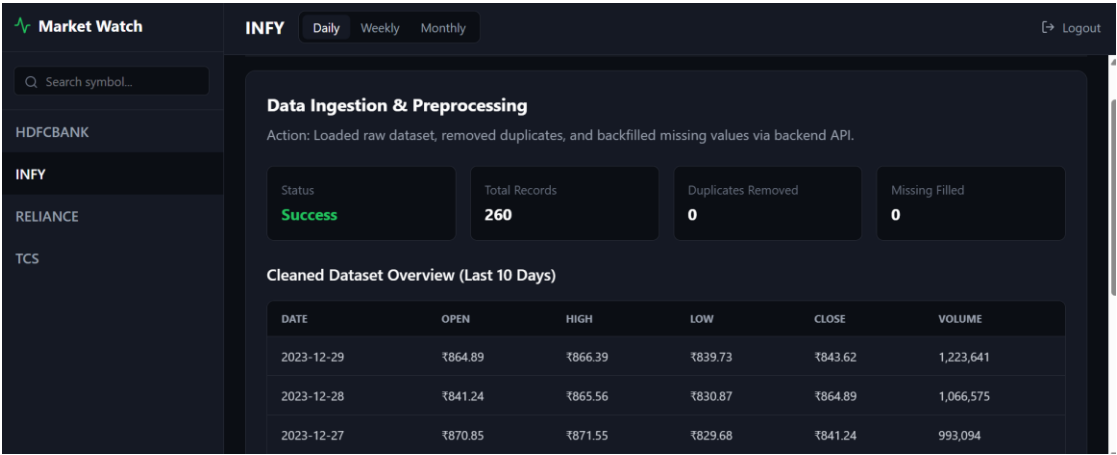


Figure 3: Preprocessing Module – Data Ingestion & Cleaning KPIs



Figure 4: Candlestick Chart and Technical Indicators Module (hover tooltip, SMA/EMA toggles)

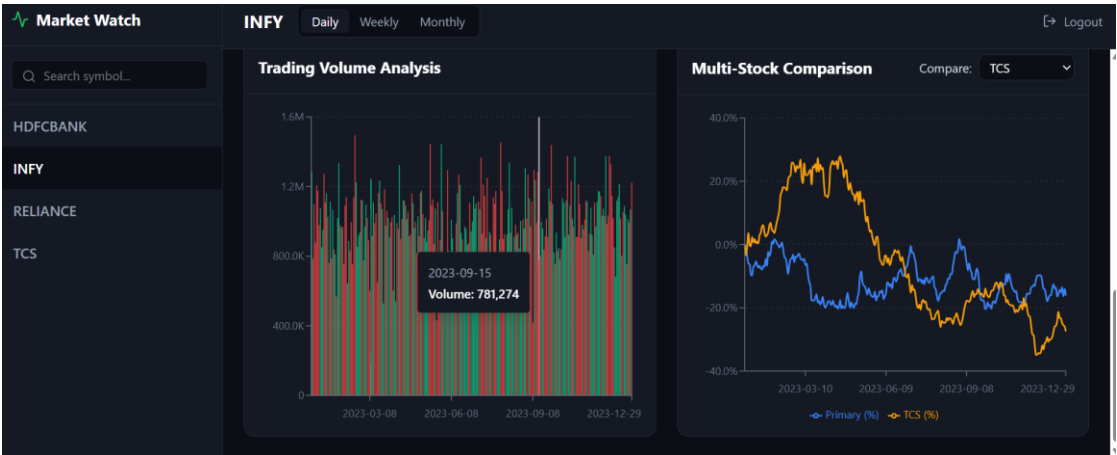


Figure 5: Trading Volume Analysis and Multi-Stock Comparison Module

Tools and Data Sources Cited

- Frontend: React, Vite, Tailwind CSS.
- Charting: TradingView lightweight-charts (candlestick rendering); Recharts (trend lines, moving-average overlays, volume bar chart).
- Backend: Python, FastAPI, pandas (DataLoader class for cleaning/preparation).
- Security: JSON Web Tokens (JWT) for authenticated API routing.
- Dataset: dummy_data.csv — mock OHLCV dataset structured to match real NSE daily price data for RELIANCE, TCS, INFY, and HDFCBANK.